Cross-Dataset Robustness of Spiral Classifiers for Parkinson's Disease Screening

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Abstract*—Handwriting impairment is a recognized manifestation of Parkinson's disease (PD), and the Archimedean spiral is a widely used, computationally simple task. Yet most machine-learning studies report accuracy on a single small dataset, leaving robustness to acquisition or population change untested. We present a 3×3 train-versus-test benchmark, run separately for each pipeline, across three datasets: the public Parkinson's Drawings dataset of Zham et al., the Folador spiral dataset, and a new longitudinal clinical dataset of 46 participants (19 controls, 27 PD) with nearly 3,500 spirals scanned over approximately one year under routine clinical conditions. Four pipelines are evaluated under one subject-level protocol: Histogram-of-Oriented-Gradients features, a fine-tuned VGG19, a hybrid CNN–NCA–SVM pipeline, and a self-supervised DINOv2 frozen-feature probe we introduce as a low-cost reference. Transfer proves target-specific rather than simply success-or-failure: models transfer between the clean public sets (DINOv2 accuracy 0.71–0.73) yet collapse to near-chance when the clinical cohort is the test set (Zham-to-clinical 0.46), degrading only partially in reverse; the wall is cohort-driven — the clinical cohort is the harder, more heterogeneous one, not milder. DINOv2 leads the in-domain ceiling on every dataset, yet reaches only 0.74 on the clinical cohort versus 0.89 on Zham. On the cross-dataset average it trails VGG19 by 2.3 pp (0.604 vs 0.627), at one quarter the configuration choices and no GPU training— comparable accuracy at lower cost given the small cohorts, not a decisive win. Strong curated-benchmark accuracy thus does not reliably predict clinical performance: deployable screening needs acquisition-robust representations and a pipeline light enough to re-fit on ordinary clinical hardware, as provided by DINOv2's frozen-backbone design.

***Index Terms*—Parkinson's disease, Archimedean spiral, cross-dataset generalization, domain shift, self-supervised learning, DINOv2, pen-and-paper data, machine learning**

I. Introduction

Parkinson's disease (PD) is the second most common neurodegenerative disorder and the fastest-growing neurological condition in terms of prevalence and associated disability worldwide [1]. According to the International Parkinson and Movement Disorder Society (MDS), PD progresses through pre-clinical, prodromal, and clinical stages, with motor impairments becoming increasingly pronounced as the disease advances [2]. By the time PD is clinically diagnosed, substantial degeneration of dopaminergic neurons in the substantia nigra has already occurred, motivating the development of objective methods capable of characterizing early disease-related motor alterations [3].

Handwriting impairments are recognized manifestations of PD and are considered in established clinical assessment instruments such as the Movement Disorders Society Unified PD Rating Scale (MDS-UPDRS), which includes handwriting as part of the evaluation of motor experiences of daily living [4]. Due to the clinical relevance of these impairments, numerous studies have explored handwriting and drawing based approaches as objective tools for PD assessment. These approaches include both online methods using digitizing tablets or smart pens to capture dynamic signals, and offline methods based on scanned pen-and-paper drawings. Online acquisition requires specialized hardware and may limit large-scale clinical adoption. Among offline approaches, evaluation is frequently performed using relatively small public datasets acquired under controlled conditions, often consisting of pre-cropped or carefully curated images [5]. The tasks investigated in the literature range from handwriting samples and specific letter sequences to geometric patterns such as loops, sinusoidal curves, concentric circles, and Archimedean spirals. These methodologies have been applied to analyze medication response, characterize motor control impairments, and develop automated diagnosis support systems [6]–[13]. Among these tasks, the Archimedean spiral has received particular attention due to its simplicity, language independence, and ability to capture motor alterations in a standardized visual format suitable for computational analysis [14]. Computerized spiral analysis, in particular, has been clinically validated as a sensitive measure in early PD [15].

While recent machine-learning advances have accelerated spiral-based PD classification [16], these methods are typically evaluated only on the small, curated datasets used for their development, leaving it unclear how well they generalize across diverse acquisition settings and populations. To address this gap, we leverage widely adopted public datasets (Zham and Folador [13], [17]) and introduce a novel large longitudinal dataset of nearly 3,500 spiral images from 46 participants (19 controls and 27 PD) performing spiral and wave drawing tasks over approximately one year, and we establish a systematic cross-dataset benchmark evaluating preprocessing and classification pipelines in every train/test direction, not within-dataset alone. We compare established methods from the literature against DINOv2 [18], a self-supervised Vision Transformer whose general-purpose features, learned from large-scale unlabeled

images, have shown strong robustness and transferability across image domains; we further trace this cross-dataset gap to cohort and label-definition differences rather than to the acquisition medium. To guarantee reproducibility, we release all training code, models, and benchmarking tools publicly[1].

II. Related Work

Table I summarizes published datasets and machine-learning approaches for PD classification from spiral drawings. Most studies rely on a few small public benchmarks — particularly the Zham Parkinson's Drawings dataset (known as Kaggle dataset with 102 spiral and 102 wave images, derived from the tablet-based cohort) [17] — alongside the Folador scanned dataset [13], merged multi-source spiral datasets [23], and proprietary or clinical cohorts [5], [19]. A recurring pattern treats the drawings as image-classification inputs, applies resizing and augmentation to offset the limited dataset size, and uses deep networks — frequently VGG and ResNet backbones — as feature extractors, either end-to-end or feeding classical classifiers such as SVM or KNN [20]–[22], [24], [25]; a common variant adds a feature-selection or hybrid stage such as Neighborhood Component Analysis [21], [24], [25], and more recent work extends the comparison to transformers such as ViT, still within the same small benchmark [22].

Overall, both handcrafted descriptors and deep representations achieve strong within-dataset performance, but direct comparison across studies is difficult because datasets, acquisition devices, preprocessing, augmentation, and evaluation protocols vary considerably, and results are typically evaluated only within the same benchmark used for development. Consequently, the cross-dataset robustness of spiral-based PD classifiers remains insufficiently characterized, motivating the present study.

III. Methods

A. Datasets

This study uses three datasets of Archimedean spiral and wave drawings from healthy controls and individuals with Parkinson's disease, all scanned paper tests. Their cohort characteristics are summarized in Table II. We explore different processing pipelines to evaluate the best option for generalization.

First, the Hemato dataset was collected by the Núcleo de Inovação e Avaliação de Tecnologias em Saúde (NIATS) at the Federal University of Uberlândia, under approval of its Research Ethics Committee (Process No. 69064623.0.0000.5152). Acquired between October 2023 and March 2025 under routine clinical conditions, it comprises nearly 3,500 scanned spirals (and a comparable number of waves), with participants repeatedly performing the tasks over approximately one year. Each visit is paired with an MDS-UPDRS-III assessment, so for the 23 PD patients seen on two or more occasions (Table II) we quantify motor change as the absolute difference in total score between their first and last visit.

[1]https://github.com/NIATS-UFU/parkinson-model-training

The other two datasets are public (Table I): the Parkinson's Drawings dataset of Zham et al. [17], distributed through Kaggle (102 spiral and 102 wave images), and the Folador dataset [13] (480 spiral and 480 wave images). For both we preserve the original Parkinson's-versus-control labels without modification, and formulate the task as binary subject-level classification across all three datasets.

B. Preprocessing

All pipelines share one evaluation protocol — identical diagnostic labels, subject identifiers, and train/test discipline — while each retains the preprocessing and augmentation of its original publication. Rather than imposing one strategy, we pair each pipeline with its native recipe: **AUG-None** (Folador [13], no augmentation), **AUG-Huang** [22], and **AUG-Dogan** [24]; full parameters are in the released code. Two underspecified Huang details were reconstructed: a "predefined range" for rotation, for which we adopt $\pm 15^{\circ}$, and per-channel normalization with mean $=$ std $= 0.5$ (Huang §2.1). Augmentations are applied only to training-fold images under a strict split-first discipline (train/test indices fixed before augmentation); test images are only resized and normalized. The DINOv2 probe is additionally evaluated across all three recipes on the Hemato within-dataset diagonal as a sensitivity check, reported under H4 in Sec. IV-C.

C. Benchmark Pipelines

Four representative pipelines were selected to cover the principal methodological families reported in the spiral-based Parkinson's disease literature. All four are trained and evaluated on the same images, labels, and splits, and each declares its number of free configuration choices together with the assumptions and reproduction decisions traceable to its source paper, so that the reproductions remain auditable against the literature. Because the source papers are frequently underspecified or internally inconsistent, we record below the choices we had to make to obtain a runnable reproduction; the full numeric settings are available in the released code.

Huang et al. [22] is reproduced as a representative end-to-end transfer-learning pipeline: an ImageNet-pretrained VGG19 fine-tuned with rotation, flipping, and cosine-annealing scheduling — twelve free configuration choices. Two assumptions were required: the source specifies only a "predefined range" for rotation, so we adopt $\pm 15^{\circ}$; and of its 6×4 backbone$\times$augmentation grid with a cosine-annealing ablation we reproduce only the single headline-best configuration, since the cross-dataset matrix, not the breadth of that in-domain grid, is our contribution.

Dogan et al. [24] is reproduced as a representative hybrid pipeline: three fine-tuned CNN backbones (DenseNet201, Xception, NASNetMobile) whose concatenated features are reduced by Neighborhood Component Analysis (NCA) and classified by a linear SVM — eleven free configuration choices. Two decisions follow: the original reports its best accuracy on a combined spiral-and-wave task, whereas we keep spirals only for consistency; and it reaches that accuracy through

TABLE I

SUMMARY OF PUBLISHED DATASETS AND MACHINE-LEARNING APPROACHES USING SPIRAL DRAWING TASKS FOR PD CLASSIFICATION.

Study	Dataset / cohort	Task and acquisition	Preprocessing / augmentation	Representation	Model / classifier	Reported performance
Seedat et al. [19]	Proprietary clinical cohort: 370 PD, 669 essential tremor, 357 controls	Scanned pen-and-paper Archimedean spirals	Resize to 224 × 224; flip, contrast, zoom, crop augmentation	CNN-learned image features	ImageNet-pretrained ResNet32	>95% accuracy
Zham et al. [17]	Original tablet-based cohort: 27 PD, 28 controls; later associated with the common Kaggle Parkinson's Drawings benchmark	Ink pen on paper over a digitizing tablet; guided Archimedean spirals	Stroke segmentation; speed and pressure extraction	Composite Index of Speed and Pen Pressure (CISP)	Statistical severity assessment	N/A (severity assessment)
Shaban [20]	Zham Parkinson's Drawings dataset: 102 spiral and 102 wave images	Spiral and wave drawings	Resize to 224 × 224; rotation augmentation	CNN transfer-learning features	Fine-tuned VGG19	89% (spiral), 88% (wave)
Varalakshmi et al. [21]	Zham Parkinson's Drawings spiral subset: 102 spiral images	Archimedean spiral drawings	Histogram equalization, resizing, augmentation	Deep CNN features	ResNet50-SVM, VGG16, VGG19, CNN-LSTM	>98% accuracy
Huang et al. [22]	Zham Parkinson's Drawings dataset: 102 spiral and 102 wave images	Spiral and wave drawings	Resize, normalization, AugMix, PixMix	Transfer-learning and transformer features	VGG16, VGG19, ResNet18/50/101, ViT	96.67% accuracy, best with VGG19
Folador et al. [13]	NIATS/Folador public dataset: 20 PD, 20 controls; 480 spirals and 480 waves	Scanned Archimedean spiral and sinusoidal drawings	Normalization to 256 × 256	Histogram of Oriented Gradients (HOG)	RF, KNN, SVM, 1D-CNN	>90% accuracy
Bernardo et al. [23]	Merged public datasets: 514 spirals from HandPD, Parkinson's Drawings, and Parkinson Disease Spiral Drawings Using Digitized Graphics Tablet	Mixed scanned and digitized Archimedean spirals	Image normalization and augmentation	SqueezeNet deep features	SVM, compared with RF and GNB	91.26% accuracy
Dogan [24]	Zham Parkinson's Drawings dataset: 102 spiral and 102 wave images	Archimedean spiral and sinusoidal wave drawings	Resize to 224 × 224; augmentation (rotation, flipping, color transforms)	CNN features refined using NCA	Hybrid CNN–NCA–SVM	99.39% accuracy
Saleh et al. [25]	Zham Parkinson's Drawings dataset: 102 spiral and 102 wave images	Spiral and wave drawings	Skeletonization, padding, augmentation	CNN features	CNN-KNN ensemble voting classifier	96.67% accuracy
Li et al. [5]	Template-free clinical dataset (Zhejiang Hospital): 43 PD, 43 controls; 166 scanned spirals	Scanned template-free Archimedean spirals	Spiral extraction, smoothing, denoising, binarization, augmentation	Continuous convolution features	CC-Net (Continuous Convolution Network)	89.3% accuracy

TABLE II

COHORT CHARACTERISTICS.

	Hemato (meas.)	**Zham**[a]	**Folador**[b]
PD / Control (n)	27 / 19	27 / 28	20 / 20
Sex PD (F/M)	12 / 15	6 / 22	3 / 17
Age PD (yr)	62.7 ± 7.8 (48–84)	71.4 ± 9.4	67.7 ± 9.1
Age Control (yr)	63.6 ± 9.4 (51–85)	71.3 ± 7.2	67.1 ± 8.4
Severity, PD	UPDRS-III 26.1 ± 14.0 (5–68)	UPDRS-III 17.6 ± 7.7	hand subscore $\sim 16/48$[c]
Hoehn & Yahr, PD[d]	1:13, 2:10, 3:4	1–1.5:12, 2–2.5:8, $\geq$3:7	n/r
Longitudinal	23 PD, $\geq$2 sessions, median 441 d span	1 session	2 sessions, $\sim$1 wk apart
Medication at drawing	ON 84%, OFF 16%	levodopa ON	levodopa ON

Hemato values are measured (REDCap and the clinical-scales workbook); Zham and Folador values are taken from the source papers. [a]Zham 2017 (literature). [b]Folador 2021 (literature). [c]**Different instrument:** a 6-item bilateral hand subscore (MDS-UPDRS items 3.4–3.6 and 3.15–3.17 — hand bradykinesia *and* tremor, summed over both hands, range 0–48), *not* the full MDS-UPDRS Part III (0–132) used by Hemato and Zham; the three severity rows therefore use different instruments and are not comparable across columns. [d]**H&Y notation:** entries are *stage:count*, i.e. the number of PD subjects at each Hoehn & Yahr stage; the Zham source reports only severity strata (1–1.5, 2–2.5, $\geq$3), so its bins are listed as published.

an exhaustive search over 129 backbone combinations (nine CNNs), of which six triples tie at 99.39% with augmentation applied before the split, so we treat the figure as an unverified upper bound and reproduce the most parameter-efficient tied triple.

Folador et al. [13] is included as a handcrafted-feature baseline: Histogram of Oriented Gradients (HOG) descriptors from grayscale 256×256 spirals fed to one of four conventional classifiers (random forest, KNN, linear SVM, 1D-CNN) — four free configuration choices. We resize to 256×256 to reproduce Folador's 8100-element feature vector; we resolve a text-vs-code disagreement on the 1D-CNN kernel size (5 against 23) in favour of the released code; and we reuse its subject-level PwPD-versus-CG labels under a Parkinson's-versus-control framing. We report the linear SVM as the headline variant and retain the other three in the supplementary material. Being largely deterministic, this pipeline offers a low-cost, reproducible reference against which the deep-learning and self-supervised pipelines can be contrasted.

Finally, we propose a novel pipeline based on DINOv2 [18]: a frozen ViT-B/14 backbone as feature extractor with a scikit-learn logistic-regression probe on its CLS-token embedding; only the probe is fit per dataset. Relying on large-scale self-supervised pretraining, it requires no fine-tuning, and with only three free configuration choices (`vit_size`, `probe_C`, `image_size`) it collapses both the configuration surface and the compute cost to a fraction of the fine-tuned CNNs. It also required none of the reproduction assumptions above — a frozen public backbone with a linear probe leaves nothing underspecified to recover — which is itself part of the reproducibility advantage examined under H4. We introduce it as a low-cost reference to test whether general-purpose self-supervised features can match task-specific supervised training while remaining far easier to reproduce.

A full per-pipeline accounting of free configuration choices — the primary, hardware-independent cost axis — is reported in Sec. IV-C, alongside the qualitative regime difference (CPU-only versus GPU); the per-cell wall-clock, GPU-memory, and

latency figures were collected on heterogeneous hardware and are not comparable, so we do not tabulate them.

D. Experimental Design

Our primary contribution is a per-pipeline 3×3 train-versus-test matrix over the Zham, Folador, and Hemato datasets, illustrated in Table III. Within this design we test three hypotheses (H1, H2, H4). The non-contiguous numbering reflects an earlier H3 — which tested whether augmentation recipes are interchangeable across pipelines — that was dropped in favour of pairing each pipeline with its native recipe for faithful reproduction, with its augmentation-sensitivity question folded into H4:

- **H1 (in-domain reproducibility ceilings):** Our reproductions reach the in-domain regime of the source papers and establish the within-dataset ceilings the cross-dataset matrix is referenced against. We expect exact parity only where the comparison is like-for-like: Huang's spiral-only single-configuration result should reproduce within ± 5 percentage points (pp), whereas Dogan's headline (combined spiral-and-wave, large backbone search) is not directly comparable to our spiral-only reproduction; we therefore report participant-grouped CV ceilings rather than claim headline parity.
- **H2 (cross-dataset generalization):** We hypothesize that transfer degrades sharply only when the clinical Hemato cohort is the target: transfer between the clean public datasets (Zham $\leftrightarrow$ Folador) should preserve subject-level accuracy, whereas transfer into Hemato from either public source loses more than 15 pp relative to the source's in-domain ceiling and falls toward chance. We read a cell as *collapsed* when subject accuracy approaches chance (≈ 0.5) and *preserved* when clearly above it, and expect the reverse direction (Hemato $\rightarrow$ public) to degrade without collapsing — a target-specific rather than uniform degradation.
- **H4 (reproducibility cost):** DINOv2 frozen + linear probe matches or beats fine-tuned CNNs cross-dataset with substantially fewer free configuration choices, lower compute, and demonstrable robustness to augmentation choice across the three literature-paired recipes.

Because every cell holds labels, protocol, and metrics fixed, performance differences trace to dataset characteristics and architecture rather than to protocol.

a) Pipeline reproduction (H1).: Each pipeline is first reproduced on its original publication's dataset and compared against the reported accuracy, so the benchmark starts from faithful reimplementations rather than re-tuned variants (the underspecified points are documented above; DINOv2 required none). Because the public datasets distribute image-level splits that do not keep each participant on one side, every within-dataset ceiling is reported under participant-grouped 5-fold cross-validation rather than those splits.

b) Cross-dataset matrix (H2).: The datasets form the matrix of Table III: off-diagonal cells train on all of one dataset and test on another with no adaptation, needing no split because the datasets hold disjoint participants; diagonal cells instead hold each participant out once across five participant-grouped folds and pool the predictions, scoring every subject exactly once (a single split would leave too few test participants for a stable estimate). Each cell reports the subject-level metric suite of Sec. III-E, with accuracy and ROC-AUC as the headline. We summarize directionality with an *asymmetry index*: the mean accuracy of the two Hemato$\rightarrow$public off-diagonals minus that of the two public$\rightarrow$Hemato off-diagonals, so a positive value marks a pipeline that transfers out of the clinical cohort better than foreign models transfer in.

TABLE III
CROSS-DATASET EVALUATION MATRIX PER PIPELINE. CROSS-DATASET (OFF-DIAGONAL) CELLS TRAIN ON ALL OF ONE DATASET AND TEST ON ANOTHER; WITHIN-DATASET (DIAGONAL) CELLS USE 5-FOLD CROSS-VALIDATION GROUPED BY PARTICIPANT.

	test on Zham	test on Folador	test on Hemato
train on Zham	in-domain	cross	cross
train on Folador	cross	in-domain	cross
train on Hemato	cross	cross	in-domain

c) Reproducibility cost (H4).: We then compare the pipelines on a second axis — reproducibility cost: the number of free configuration choices and the amount of computation each requires. The Hemato within-dataset diagonal in particular shows how every method behaves on routine clinical acquisitions and provides the common, like-for-like basis for this comparison (H4).

E. Evaluation Metrics

Every cell is scored with the same suite, so pipelines and datasets remain directly comparable. The task is binary classification of Parkinson's disease against control: each model produces a positive-class probability per image, thresholded at 0.5 to a predicted label, from which we compute accuracy, sensitivity (recall), specificity, precision, F_1-score, and the area under the ROC curve (ROC-AUC).

All metrics are reported at the *subject* level, since the clinical decision is per patient: a subject's per-image positive-class probabilities are averaged and the mean thresholded at 0.5. For Zham, subject identity is recovered from image filenames rather than a published participant key, but most Zham subjects contribute a single image, so its subject-level aggregation largely coincides with per-image scoring. Subject-level accuracy is the headline metric, reported with ROC-AUC and percentile bootstrap 95% confidence intervals over 1000 resamples.

ROC-AUC is reported wherever defined. Because it is undefined when a fold contains a single class, the within-dataset diagonal pools the out-of-fold predictions across all five folds and computes a single ROC-AUC over them (every subject scored exactly once) rather than averaging per-fold values; the remaining metrics follow scikit-learn's convention of returning zero when the class needed for their denominator is absent from a fold.

IV. Results

A. In-domain reproducibility and ceilings (H1)

We first situate the reproductions against the source papers (Table IV). Only Huang's spiral configuration admits a like-for-like comparison: its reproduction on the released Zham split (0.840) lands within ±5 pp of the 0.87 spiral accuracy Huang reports (the widely cited 96.67% is the *wave* result) [22]. Dogan's headline 99.39% was obtained on a *combined* spiral-and-wave task over a 129-combination backbone search whose augmentation is applied before the train/test split, which leads us to treat it as an unverified upper bound; our spiral-only single-configuration reproduction reaches 0.760 on the same released split, so we do not claim headline parity. Folador report 0.696 for the spiral SVM under 5-fold CV with an unspecified fold rule [13], precluding fixed-split parity; our runs instead *bracket* it — participant-grouped CV below (0.575), the shipped single split far above (0.950) — a spread implying neither that split nor Folador's own folds are participant-disjoint (Sec. III-D). The contrast between the released-split column and the grouped-CV ceiling column of Table IV is why the cross-dataset matrix is referenced against the grouped-CV ceilings rather than released-split numbers.

TABLE IV
H1 REPRODUCTION FIDELITY. *Published* = THE SOURCE PAPER'S ACCURACY UNDER ITS OWN PROTOCOL; *released-split* = OUR REPRODUCTION ON THE PUBLICLY DISTRIBUTED TRAIN/TEST SPLIT; *ceiling* = OUR POOLED-OOF 5-FOLD PARTICIPANT-GROUPED CV ACCURACY. EACH PIPELINE IS SHOWN ON ITS ORIGINAL DATASET.

Pipeline (dataset)	Published	Released-split	Grouped-CV ceiling
Huang VGG19 (Zham)	0.87[a]	0.840	0.786
Dogan triple (Zham)	0.994[b]	0.760	0.786
Folador HOG-SVM (Folador)	0.696[c]	0.950	0.575

[a]spiral, single split. [b]spiral+wave combined, single split (upper bound). [c]spiral, 5-fold CV.

In-domain ceilings are therefore reported as pooled out-of-fold subject accuracy under 5-fold participant-grouped cross-validation, each with ROC-AUC and a percentile bootstrap 95% confidence interval (Table V). DINOv2 (frozen) attains the highest ceiling on Zham (0.893) and Folador (0.875) and also leads on Hemato (0.739); among the fine-tuned CNNs, Huang VGG19 (AUG-Huang) leads on Folador (0.850). On Hemato all four pipelines cluster in the 0.67–0.74 band, confirming that the clinical cohort is hard for every pipeline even in-domain.

B. Cross-dataset generalization (H2)

Fig. 1 reports the four cross-dataset matrices. Zham-trained and Folador-trained probes drop to near-chance subject accuracy when tested on Hemato (DINOv2 AUG-None: Z→H 0.457, F→H 0.522), while Hemato-trained probes degrade more gracefully on the public datasets (H→Z 0.679, H→F 0.525). Cross-public transfer between the two clean datasets does *not* collapse: DINOv2 AUG-None reaches Z→F 0.725 and F→Z 0.714. The generalization wall is therefore target-specific (→Hemato) and cohort-driven, not a uniform clinical-versus-clean asymmetry.

TABLE V
IN-DOMAIN CEILINGS: POOLED-OOF 5-FOLD SUBJECT-CV SUBJECT ACCURACY WITH PERCENTILE BOOTSTRAP 95% CI AND POOLED ROC-AUC. EACH PIPELINE UNDER ITS NATIVE RECIPE.

Pipeline (recipe)	Dataset	Subj. acc. [95% CI]	AUC
DINOv2 (AUG-None)	Zham	**0.893** [0.785–1.000]	0.949
DINOv2 (AUG-None)	Folador	**0.875** [0.775–0.975]	0.892
DINOv2 (AUG-None)	Hemato	**0.739** [0.609–0.848]	0.752
Folador HOG (AUG-None)	Zham	0.714 [0.536–0.893]	0.841
Folador HOG (AUG-None)	Folador	0.575 [0.400–0.725]	0.693
Folador HOG (AUG-None)	Hemato	0.674 [0.522–0.804]	0.686
Huang VGG19 (AUG-Huang)	Zham	0.786 [0.643–0.929]	0.877
Huang VGG19 (AUG-Huang)	Folador	0.850 [0.750–0.950]	0.920
Huang VGG19 (AUG-Huang)	Hemato	0.674 [0.543–0.804]	0.789
Dogan triple (AUG-Dogan)	Zham	0.786 [0.643–0.929]	0.800
Dogan triple (AUG-Dogan)	Folador	0.775 [0.650–0.900]	0.922
Dogan triple (AUG-Dogan)	Hemato	0.717 [0.565–0.848]	0.719

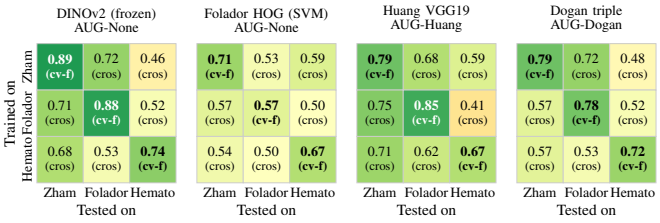


Fig. 1. Cross-dataset subject accuracy per pipeline under the locked protocol (off-diagonals: full-A → full-B; diagonals: pooled 5-fold subject CV). Models trained on Zham or Folador collapse to chance on Hemato; models trained on Hemato degrade but do not collapse on the public datasets (H→Z ≈ 0.68 and H→F ≈ 0.53 for DINOv2 AUG-None). The wall is target-specific (→Hemato) rather than symmetric.

The same broad pattern recurs across the pipelines (Table VI, Fig. 2), with pipeline-dependent strength. Huang VGG19 (AUG-Huang) actually leads the off-diagonal mean (0.627) while DINOv2 AUG-None sits at 0.604 — a 2.3 pp gap — but at one quarter the configuration cost (Sec. IV-C). Hemato→Zham is not uniformly high across pipelines under the locked protocol; it ranges 0.536 (Folador) to 0.714 (Huang).

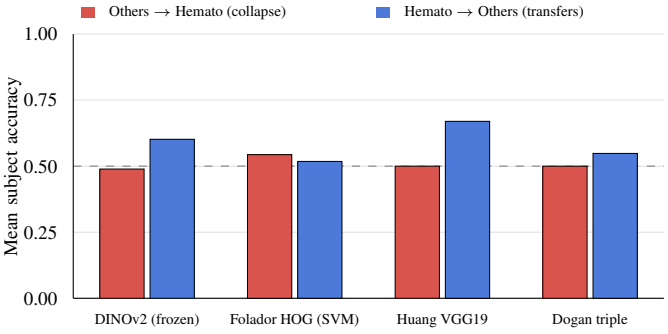


Fig. 2. Per-pipeline asymmetry index: mean subject accuracy of H→{K,F} minus mean of {K,F}→H. Three of the four pipelines show a positive asymmetry; Folador HOG is slightly negative, as its into-Hemato cells degenerate to majority-class (all-Parkinson's) prediction. The sign and magnitude are therefore pipeline-dependent rather than uniform.

a) Folador as a test target.: Folador is anomalous in two ways. First, its in-domain HOG ceiling is at chance (0.575 pooled-CV; Table V) despite a strong released-split

TABLE VI

CROSS-DATASET OFF-DIAGONAL SUBJECT-LEVEL METRIC SUITE, EACH PIPELINE UNDER ITS NATIVE RECIPE. Z = ZHAM, F = FOLADOR, H = HEMATO; DIRECTION A→B TRAINS ON ALL OF A AND TESTS ON ALL OF B (LOCKED PROTOCOL). ACCURACY CARRIES A PERCENTILE BOOTSTRAP 95% CI. OFF-DIAGONAL MEAN ACCURACY IS REPORTED IN THE LAST ROW OF EACH BLOCK.

Pipeline	Dir	Acc [95% CI]	AUC	Sens	Spec	Prec	F_1
DINOv2 AUG-None	Z→F	0.725 [0.60–0.88]	0.843	0.55	0.90	0.85	0.67
DINOv2 AUG-None	Z→H	0.457 [0.33–0.59]	0.575	0.15	0.90	0.67	0.24
DINOv2 AUG-None	F→Z	0.714 [0.54–0.86]	0.918	0.47	1.00	1.00	0.64
DINOv2 AUG-None	F→H	0.522 [0.39–0.65]	0.649	0.33	0.79	0.69	0.45
DINOv2 AUG-None	H→Z	0.679 [0.50–0.82]	0.805	0.80	0.54	0.67	0.73
DINOv2 AUG-None	H→F	0.525 [0.38–0.68]	0.545	0.90	0.15	0.51	0.66
DINOv2 AUG-None	*mean*	**0.604**	—	—	—	—	—
Huang AUG-Huang	Z→F	0.675 [0.53–0.83]	0.814	1.00	0.35	0.61	0.76
Huang AUG-Huang	Z→H	0.587 [0.44–0.72]	0.426	1.00	0.00	0.59	0.74
Huang AUG-Huang	F→Z	0.750 [0.61–0.89]	0.872	0.60	0.92	0.90	0.72
Huang AUG-Huang	F→H	0.413 [0.28–0.57]	0.591	0.00	1.00	0.00	0.00
Huang AUG-Huang	H→Z	0.714 [0.54–0.86]	0.836	0.67	0.77	0.77	0.71
Huang AUG-Huang	H→F	0.625 [0.48–0.78]	0.670	1.00	0.25	0.57	0.73
Huang AUG-Huang	*mean*	**0.627**	—	—	—	—	—
Dogan AUG-Dogan	Z→F	0.725 [0.58–0.85]	0.895	0.80	0.65	0.70	0.74
Dogan AUG-Dogan	Z→H	0.478 [0.33–0.63]	0.480	0.59	0.32	0.55	0.57
Dogan AUG-Dogan	F→Z	0.571 [0.39–0.75]	0.697	1.00	0.08	0.56	0.71
Dogan AUG-Dogan	F→H	0.522 [0.37–0.65]	0.505	0.63	0.37	0.59	0.61
Dogan AUG-Dogan	H→Z	0.571 [0.39–0.75]	0.605	0.20	1.00	1.00	0.33
Dogan AUG-Dogan	H→F	0.525 [0.38–0.68]	0.510	0.35	0.70	0.54	0.42
Dogan AUG-Dogan	*mean*	0.565	—	—	—	—	—
Folador HOG AUG-None	Z→F	0.525 [0.38–0.68]	0.485	1.00	0.05	0.51	0.68
Folador HOG AUG-None	Z→H	0.587 [0.44–0.72]	0.561	1.00	0.00	0.59	0.74
Folador HOG AUG-None	F→Z	0.571 [0.39–0.75]	0.559	0.93	0.15	0.56	0.70
Folador HOG AUG-None	F→H	0.500 [0.37–0.63]	0.624	0.30	0.79	0.67	0.41
Folador HOG AUG-None	H→Z	0.536 [0.36–0.71]	0.682	1.00	0.00	0.54	0.70
Folador HOG AUG-None	H→F	0.500 [0.35–0.65]	0.722	1.00	0.00	0.50	0.67
Folador HOG AUG-None	*mean*	0.537	—	—	—	—	—

number — consistent with that split not being participant-disjoint (Sec. III-D) and with spirals being the weaker modality in Folador's own analysis [13]. Second, when Folador is the *target* of transfer (columns Z→F and H→F of Table VI), foreign-trained models over-predict Parkinson's: sensitivity is high (≥ 0.90 for Folador HOG and Huang) while specificity collapses, strongest *into* Folador from Hemato (specificity 0.00–0.25) and mildest for Dogan (0.65–0.70). A plausible cause is that, although its labels are subject-level PwPD-vs-CG like ours, Folador frames them as a tremor proxy on a mild cohort — some patients show no tremor in a session and some controls show anxiety tremor (§2.6 of [13]) — introducing label noise that a foreign PD-vs-HC model registers as a specificity drop. Folador is therefore not intrinsically weak but encodes a different labeling target that our framing exposes.

C. Reproducibility cost (H4)

The hardware-independent cost axis is the number of free configuration choices and the training regime each pipeline requires (Table VII). DINOv2 reaches the highest Hemato pooled-OOF subject accuracy (0.739) with only three configuration choices and no fine-tuning — a frozen backbone whose features are extracted in a single pass, with one logistic probe fit per fold. The two fine-tuned CNNs reach equal or lower accuracy at far higher configuration cost: Dogan triple (0.717) exposes eleven choices and Huang VGG19 (0.674) twelve, both with full backbone fine-tuning. The remaining pipeline, the deterministic Folador HOG, matches Huang (0.674) with only four configuration choices and no fine-tuning, yet still trails DINOv2 by 6.5 pp.

TABLE VII

REPRODUCIBILITY COST: FREE CONFIGURATION-CHOICE COUNT, TRAINING REGIME, AND POOLED-OOF HEMATO 5-FOLD CV SUBJECT ACCURACY PER PIPELINE UNDER ITS NATIVE RECIPE (PER-FOLD σ IN PARENTHESES).

Pipeline (recipe)	# cfg	Training regime	Hemato CV
DINOv2 (frozen) AUG-None	3	frozen backbone; single-pass features + 1 logistic fit	**0.739** (σ 0.115)
Folador HOG AUG-None	4	deterministic HOG + classifier fit (no epochs)	0.674 (σ 0.158)
Dogan triple AUG-Dogan	11	3 backbones fine-tuned + NCA + linear SVM	0.717 (σ 0.124)
Huang VGG19 AUG-Huang	12	end-to-end fine-tune, 50 epochs	0.674 (σ 0.100)

Because the four pipelines were run on heterogeneous hardware, wall-clock, latency, and GPU-memory figures are not directly comparable, and a uniform-hardware re-run is left to future work. The qualitative regime difference is nonetheless clear from the training regimes in Table VII: DINOv2 and Folador require no GPU at all, whereas the fine-tuned CNNs do.

Beyond configuration count, the three free DINOv2 choices produce stable results across the augmentation surface: Hemato pooled-OOF accuracy moves only from 0.739 (AUG-None) to 0.761 (AUG-Huang) to 0.717 (AUG-Dogan), a ~ 4 pp swing, so the practitioner need not search recipes to reach competitive accuracy — the strong form of the H4 reproducibility-cost argument.

Across the six matched off-diagonal cells (Table VI), DINOv2's mean subject accuracy (0.604) trails Huang VGG19's (0.627) by 2.3 pp while the two trade per-cell wins, so DINOv2 matches the best CNN at one quarter the configuration surface and still leads the in-domain ceilings on all three datasets

(Table V).

V. Discussion

a) Asymmetric cross-dataset generalization.: The spiral-PD literature largely treats cross-dataset transfer as a binary "works / doesn't work" question. Our matrix shows the wall is *target-specific*: cross-public transfer between the two clean datasets (Zham ↔ Folador) preserves accuracy, transfer into the clinical Hemato cohort from either public source collapses, and the converse direction (Hemato → public) degrades but does not collapse uniformly — Hemato→Zham works for Huang (0.714) but is weak for Folador (0.536). The relevant distinction is between *clean public scans* (Zham, Folador) and our *clinical-cohort* acquisition (Hemato), not a uniform clinical-vs-clean asymmetry. Implication: in-domain accuracy on Zham is not a useful ceiling for clinical deployment. Whether this wall yields to greater data *diversity* and different training strategies, rather than scale alone, is an open question this benchmark raises but cannot yet settle: such diversity has been reported to aid out-of-distribution robustness in adjacent domains like ECG and EEG [26].

b) Where the gap comes from.: Folador and Hemato are both pen-and-paper scanned, so the wall is not explained by acquisition medium alone: Zham and Folador form one transfer family (public clean scans), while Hemato is the independent clinical acquisition that exposes the real generalization gap. The into-Hemato wall is driven primarily by cohort and label-definition differences. Contrary to the intuition that the clinical cohort might simply be milder, Hemato is the *harder* cohort: more motor-impaired (MDS-UPDRS-III 26.1 vs Zham's 17.6), wider in severity range (5–68), younger, and far more heterogeneous than the curated public sets (Table II). Its longitudinal design deepens that heterogeneity, since one fixed binary label spans each patient's shifting motor state — a mean within-subject UPDRS-III swing of 7.3 points (max 27) — inflating within-class spread in a way the single-session public sets cannot. Folador adds a different mismatch: its labels are a tremor-presence proxy rather than a clinical diagnosis (§2.6 of [13]). Medication state, by contrast, is *not* such an axis, since Hemato spirals are drawn 84% levodopa-ON like both public cohorts. A second, minor factor is data quality: 4.9% of Hemato spirals (173/3495) are clipped where a drawing overflows its grid cell — too small to explain the collapse. We do not fully isolate the mechanism: future work could decompose the residual gap in latent space (e.g. Wasserstein distance and t-SNE on the embeddings [26]) and via saliency on into-Hemato errors. A complementary remedy is gated by data: each matrix cell trains on a single source, whereas narrowing the wall likely calls for training across cohorts at once — progress that will need the community to release more public spiral datasets.

c) Reproducibility cost as a deployment criterion.: H4 is framed not just as "DINOv2 generalizes well" but as "DINOv2 is the only pipeline that pairs a clinical-compatible cost footprint — few configuration choices, no GPU training — with competitive cross-dataset accuracy." Frozen weights mean no training divergence, and the three free choices (`vit_size`, `probe_C`, `image_size`) can be selected on a small validation split without the multi-day search Huang's twelve would imply — which, for clinical translation, matters as much as the accuracy.

d) Subject identity in the public splits.: Our subject-level framing presumes a dependable image-to-participant mapping; this holds for Hemato and Folador but is weaker for Zham, whose participant key we reconstruct from filenames rather than a published identifier. Because most Zham subjects contribute a single drawing, its subject-level scores stay close to per-image scores, and any Zham subject-level property — including the cross-validation behind its in-domain ceiling — is only as reliable as those identifiers; we therefore read the Zham in-domain numbers as a soft ceiling and rest the cross-cohort claims on Folador and Hemato, where participant identity is explicit.

e) Limitations.: Several constraints bound our conclusions. First, all three cohorts are small (40–55 subjects), so the bootstrap confidence intervals are wide and overlap heavily (Table VI): most between-pipeline differences, including DINOv2's 2.3 pp gap behind Huang, are not significance-tested and our rankings are indicative point-estimate orderings. Second, the literature pipelines are bounded reimplementations: the source papers are underspecified at several points (Sec. III-D) and we reproduce a single headline configuration rather than each study's full search, so our in-domain figures—notably Dogan's 0.760 against its 0.994 headline—are lower bounds, and the cross-dataset results characterise these reproductions, not the published models. Third, the reproducibility-cost argument rests on the configuration-choice count, since the wall-clock, latency, and GPU-memory figures were collected on heterogeneous hardware and are not comparable. Fourth, the benchmark uses spirals only, leaving the wave modality and spiral–wave fusion unexplored—which also makes our spiral-only Dogan reproduction not directly comparable to its combined-modality headline. Fifth, we attribute the into-Hemato wall to cohort and label-definition differences but do not fully decompose the residual contribution of acquisition hardware and stroke-level morphology, nor confirm that multi-source training closes the gap, which is left to future work. Finally, the matrix spans a single clinical cohort (Hemato, single-site UFU) and a binary PD-versus-control task, so we cannot tell whether the wall is general or a peculiarity of Hemato; progression regression against MDS-UPDRS deltas and the ON/OFF medication-state effect are out of scope. Resolving these will require additional clinical cohorts and multi-site replication.

VI. Conclusion

We presented a 3×3 cross-dataset robustness benchmark of pen-and-paper Archimedean spiral classifiers for Parkinson's disease, comparing three reproduced literature pipelines — the Folador HOG family [13], the Huang VGG19 fine-tune [22], and the Dogan triple [24] — with a self-supervised DINOv2 frozen-feature probe [18] introduced as a low-cost reference,

across two public datasets and a new longitudinal clinical cohort of 46 participants (19 controls, 27 patients). Transfer is target-specific rather than simply success-or-failure: models transfer well between the clean public datasets but fall to near-chance whenever the clinical Hemato cohort is the test set, degrading only partially in reverse — a cohort-driven wall, not a uniform clean-versus-clinical divide, since Hemato is hard even in-domain (subject accuracy 0.67–0.74). The limiting factor for clinical translation is thus not in-domain accuracy, already saturated in the literature, but generalization across acquisition settings. The DINOv2 probe is competitive throughout: it holds the highest in-domain ceiling on every dataset and comes within 2.3 pp of Huang's VGG19 on the cross-dataset average (0.604 versus 0.627), at one quarter the configuration choices and no fine-tuning. With only 40–55 subjects per cohort, the overlapping confidence intervals support comparable accuracy at lower cost rather than a decisive win, and its frozen-backbone design can be re-fit on a clinic's own data without GPU training. Confirming whether the wall is general or particular to Hemato will require additional clinical cohorts and multi-site replication, while narrowing it points to acquisition-invariant or domain-adaptation methods, including multi-source training across cohorts, and to the wave modality, spiral–wave fusion, and progression estimation already enabled by Hemato.

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